

Package ‘ANUBIX’

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Title Accurate Network Enrichment Analysis Test

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Description Includes examples with data for accurate network enrichment analysis of gene sets against pathways.

Depends R (>= 3.4.0)

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anubix

*Perform ANUBIX for the provided gene sets***Description**

Computes ANUBIX, an accurate test for network enrichment analysis between query sets and pathway sets. Instead of normal random sampling it does constrained random sampling, taking the degree of the nodes into account.

Usage

```
anubix(network, links_matrix, genesets, pathways, cores = 2, cutoff = 0.8,
        sampling = 2000, network_type = "weighted")
```

Arguments

network	A data.frame. Two columns if the network has no weights, where column 1 and column 2 are genes. Each row means a link between genes. If the network is weighted, then the third column contains the weights of the links.
links_matrix	A numeric matrix. Stores the links each gene has to each pathway. The number of rows equals the number of genes in the network and the number of columns equals the number of pathways under study.
genesets	A data.frame of two columns. Column 1 contains the genes and column 2 the experiment where they belong.
pathways	A data.frame of two columns. Column 1 contains the genes and column 2 the pathway where they belong.
cores	A numeric value. Number of cores used by the algorithm. Default is 2 .
cutoff	A numeric value. Link confidence threshold of the weights between genes. Default is 0.8 .
sampling	A numeric value. Number of random samplings required to construct the null distribution. Default is 2000 .
network_type	Either "weighted" or "unweighted". Default is weighted .

Value

A data frame with the following columns:

- geneset - Gene set under study.
- pathway - Pathway under study.
- obv_links - Observed number of links between the gene set and the pathway.
- exp_mean - Expected number of links between the gene set and the pathway.
- overlap - Number of genes shared by the gene set and the pathway.
- p-value - p-value of the test.
- q-value - Corrected p-value using Benjamini-Hochberg.
- FWER - Corrected p-value using Bonferroni correction.

Note

anubix_links() needs to be run beforehand

See Also

[anubix_links](#), [anubix_transitivity](#), [anubix_clustering](#), [example_anubix](#)

Examples

```
## Not run:
anubix(
  network = example_anubix$network,
  links_matrix = example_anubix$links_genes,
  genesets = example_anubix$gene_set,
  pathways = example_anubix$pathway_set,
  cores = 2,
  cutoff = 0.8,
  sampling = 2000,
  network_type = "weighted"
)

## End(Not run)
```

anubix_clustering

Network clustering of the gene set and then ANUBIX is applied

Description

Clusters the gene set using Infomap and then applies ANUBIX.

Usage

```
anubix_clustering(network, links_matrix, genesets, pathways, cores = 2,
  sampling = 2000)
```

Arguments

network	A data.frame. Two columns if the network has no weights. Where the first column and the second column are genes. Each row means a link between genes. If the network is weighted, then the third column are the weights of the links.
links_matrix	A numeric matrix. links_matrix stores the links each gene has to each pathway. The number of rows equals the number of genes in the network and the number of columns equals the number of pathways under study.
genesets	A data.frame of two columns. The first column contains the genes and the second column the experiment where they belong to.
pathways	A data.frame of two columns. The first column contains the genes and the second column the pathway where they belong to.
cores	A numeric value. Cores is defined as the number of cores used by the algorithm. The default value is 2
sampling	A numeric value. Sampling is defined as the number of random samplings required to construct the null distribution. The default value is 2000

Value

A list with two dataframes:

- geneset - Gene set under study.
- pathway - Pathway under study.
- obv_links - Observed number of links between the gene set and the pathway.
- exp_mean - Expected number of links between the gene set and the pathway.
- overlap - Number of genes shared by the gene set and the pathway.
- p-value - p-value of the test.
- q-value - Corrected p-value using Benjamini-Hochberg.
- FWER - Corrected p-value using Bonferroni correction.
- module - Module from the clustering approach.

Note

anubix_links() needs to be run beforehand

See Also

[anubix](#), [anubix_transitivity](#), [anubix_links](#), [example_anubix](#)

Examples

```
# We provide with example data to be able to run ANUBIX.
## Not run:
anubix_clustering(network = example_anubix$network,
                  links_matrix = example_anubix$links_genes,
                  genesets = example_anubix$gene_set,
                  pathways = example_anubix$pathway_set,
                  cores = 2,
                  sampling = 2000)

## End(Not run)
```

anubix_links

Create the Link Matrix Between Genes and Pathways

Description

Computes the number or strength of connections that each gene in a network has to each pathway. This matrix can be used for downstream network enrichment analysis.

Usage

```
anubix_links(network, cores = 2, pathways, cutoff = 0.8, network_type = "weighted")
```

Arguments

network	A data.frame representing the gene network. • If unweighted: Two columns (gene1, gene2), where each row represents an edge. • If weighted: Three columns, with the third column containing edge weights.
pathways	A data.frame with two columns: • Column 1: gene names. • Column 2: corresponding pathway names.
cutoff	Numeric. Threshold to filter edges in a weighted network by confidence. Default is 0.8 .
network_type	Character. Either "weighted" or "unweighted". Default is " weighted ".
cores	Integer. Number of CPU cores to use for parallel computation. Default is 2.

Value

A numeric matrix where:

- Rows correspond to genes in the network.
- Columns correspond to pathways.
- Entries represent the number (or sum of weights) of links from each gene to each pathway.

See Also

[anubix](#), [anubix_transitivity](#), [example_anubix](#), [anubix_clustering](#)

Examples

```
## Not run:
# Example run on provided example data
anubix_links(
  network = example_anubix$network,
  pathways = example_anubix$pathway_set,
  cutoff = 0.8,
  network_type = "weighted"
)

## End(Not run)
```

anubix_transitivity	<i>Perform ANUBIX transitivity test for the provided gene sets</i>
---------------------	--

Description

Computes ANUBIX transitivity, an accurate test for network enrichment analysis between query sets and pathway sets. Instead of normal random sampling, it performs constrained random sampling, taking the degree of the nodes into account. Additionally, it incorporates the gene set's transitivity to compute enrichment.

Usage

```
anubix_transitivity(network, links_matrix, genesets, pathways, cores = 2, cutoff = 0.8,
  sampling = 2000, network_type = "weighted")
```

Arguments

network	A data.frame. Two columns if the network has no weights, where the first column and second column are genes. Each row represents a link between genes. If weighted, the third column contains the weights of the links.
links_matrix	A numeric matrix. Stores the links each gene has to each pathway. Rows correspond to genes in the network; columns correspond to pathways.
genesets	A data.frame with two columns: the first column contains genes, the second column the experiment they belong to.
pathways	A data.frame with two columns: the first column contains genes, the second column the pathway they belong to.
cores	Numeric. Number of CPU cores used by the algorithm. Default is 2 .
cutoff	Numeric. Link confidence threshold for weights between genes. Default is 0.8 .
sampling	Numeric. Number of random samplings to construct the null distribution. Default is 2000 .
network_type	Character. Either "weighted" or "unweighted". Default is weighted .

Value

A data.frame with columns:

- geneset - Gene set under study.
- pathway - Pathway under study.
- obv_links - Observed number of links between the gene set and the pathway.
- exp_mean - Expected number of links between the gene set and the pathway.
- overlap - Number of genes shared by the gene set and the pathway.
- p-value - p-value of the test.
- q-value - Corrected p-value using Benjamini-Hochberg procedure.
- FWER - Corrected p-value using Bonferroni correction.

Note

anubix_links() needs to be run beforehand.

See Also

[anubix_links](#), [example_anubix](#), [anubix_clustering](#), [anubix](#)

Examples

```
## Not run:
anubix_transitivity(
  network = example_anubix$network,
  links_matrix = example_anubix$links_genes,
  genesets = example_anubix$gene_set,
  pathways = example_anubix$pathway_set,
```

```

cores = 2,
cutoff = 0.8,
sampling = 2000,
network_type = "weighted"
)

## End(Not run)

```

example_anubix

Example Data Provided with the Package

Description

A list containing example data for running ANUBIX, including a precomputed gene-pathway link matrix, pathway definitions, a gene set, and a network subset.

The list contains:

- **links_genes** – A gene-pathway link matrix computed using a confidence cutoff of 0.8 in the functional association network FunCoup 5.
- **pathway_set** – A data.frame of 288 KEGG pathways for *Homo sapiens*.
- **gene_set** – An example gene set consisting of 42 genes.
- **network** – A subset of the FunCoup 5 network for *Homo sapiens*, containing 2000 links and 1788 genes.

Usage

```
example_anubix
```

Format

An object of class list of length 4.

Source

- **pathway_set:** Kanehisa, M., and Goto, S. (2002). KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Research*, 28(1), 27–30.
- **network:** Persson, E., Castresana-Aguirre, M., Buzzao, D., Guala, D., & Sonnhammer, E. L. L. (2021). FunCoup 5: functional association networks in all domains of life, supporting directed links and tissue-specificity. *Journal of Molecular Biology*, 433(11), 166835. doi: [10.1016/j.jmb.2021.166835](https://doi.org/10.1016/j.jmb.2021.166835)

References

- Kanehisa, M., and Goto, S. (2002). KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Research*, 28(1), 27–30.
- Persson, E., Castresana-Aguirre, M., Buzzao, D., Guala, D., & Sonnhammer, E. L. L. (2021). FunCoup 5: functional association networks in all domains of life, supporting directed links and tissue-specificity. *Journal of Molecular Biology*, 433(11), 166835. doi: [10.1016/j.jmb.2021.166835](https://doi.org/10.1016/j.jmb.2021.166835)

See Also

[anubix](#), [anubix_transitivity](#), [anubix_links](#), [anubix_clustering](#)

Examples

```
## Not run:
data(example_anubix)

# We provide the precomputed link matrix (FunCoup 5, cutoff = 0.8) to run ANUBIX directly.

# Alternatively, compute the link matrix from the example network:
anubix_links(
  network = example_anubix$network,
  pathways = example_anubix$pathway_set,
  cutoff = 0.75,
  network_type = "weighted"
)

# Run ANUBIX using the example data:
anubix(
  network = example_anubix$network,
  links_matrix = example_anubix$links_genes,
  genesets = example_anubix$gene_set,
  pathways = example_anubix$pathway_set,
  cores = 2,
  cutoff = 0.8,
  sampling = 2000
)

## End(Not run)
```

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